

Use of pocket calculators is permitted, and of the formula sheet. You have 25 minutes to work on all the problems.

Academic integrity requires that you neither give nor receive any aid on this exam.

Grading policy:	Exercise	Points	Score
	1	35	
	2	35	
	3	35	
	Maximum possible	100	

Exercise 1. Consider the matrices $A = \begin{pmatrix} 1 & -2 \\ 0 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 4 \\ -1 & 0 \end{pmatrix}$.

- (a) Compute $A + B$ and AB .
- (b) Is the matrix AB symmetric? Justify your answer.
- (c) Find the inverse of AB if possible.

Exercise 2. Consider the system of equations:

$$\begin{cases} x + 2y + 3z &= 1 \\ 2x - y + z &= 2 \\ x - 2y + 2z &= -1 \end{cases} \quad (1)$$

- (a) Write the system in matrix form $A\vec{x} = \vec{b}$.
- (b) Determine the rank of the coefficient matrix A and whether the system has a unique solution, infinitely many solutions, or no solution without solving the system.
- (c) If possible, use Cramer's method to determine the value of y in the solution.

Exercise 3. Let $f_1(x, y, z, t) = x^2 - y^2 + zt = -3$ and $f_2(x, y, z, t) = e^x + \frac{y}{z} + t^2 = 0$ define a system of equations $\vec{f}(x, y, z, t) = \vec{c}$ and consider the point $P = (0, -2, 1, 1)$.

- (a) Verify if P satisfies the relations imposed by the system.
- (b) Determine if y can be expressed as a function of x near P .
- (c) Compute $(\frac{\partial y}{\partial t})_x$ at P .